AKASHDIP MAHAPATRA
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PROFESSIONAL SUMMARY

Results-driven Systems Engineer with a strong foundation in mechanical systems thinking, **currently managing enterprise-scale AWS serverless architectures and CI/CD pipelines.** Proven ability to dive deep into **Linux fundamentals**, **network troubleshooting**, and **cloud automation**. Passionate about operational excellence, infrastructure reliability, and building highly available distributed systems.

CORE COMPETENCIES

- **Cloud Infrastructure:** AWS (Lambda, ECR, ECS, S3, IAM, VPC, EventBridge, MSK/Kafka, EC2, DynamoDB), & fundamentals of Azure (Azure DevOps) and Google Cloud Platform.
- **DevOps & CI/CD:** GitHub Actions, Docker, Jenkins, Terraform, Infrastructure as Code (IaC), Advanced Git Version Control (manual rebasing, remote syncs).
- **Operating Systems & Networking:** Linux (Ubuntu, RHEL), TCP/IP, DNS, HTTP/HTTPS, Nginx, Wireshark, nmap.
- **Programming & Scripting:** Python (Boto3, Data Structures & Algorithms), Bash, C, SQL, Oracle SQL.
- **Security & Observability:** Vulnerability Management (CVE patching), CloudWatch, Datadog, IAM Least Privilege.

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS) | Onsite *Data Engineer (DevOps & Infrastructure Focus under British Airways)* | Nov 2025 - Present

- Manage and Automated CI/CD deployment lifecycle (GitHub Actions to AWS ECR) for a multi-tiered AWS serverless data pipeline processing large-scale IoT and telemetry data.
- Support and troubleshoot a complex event-driven architecture utilizing Amazon MSK (Kafka), EventBridge, and containerized AWS Lambda functions to ensure high availability.
- Perform Root Cause Analysis (RCA) to identify, investigate, and remediate critical security vulnerabilities (CVEs) within Docker containers and Python dependencies, successfully reducing image sizes and hardening production security.
- Engineered automated post-deployment validation workflows for the GSE Events Data Product using Python, Boto3, and GitHub Actions, integrating CloudWatch and Datadog APIs to eliminate manual log-checking and accelerate deployment sign-offs.
- Successfully architected and onboarded the Pathfinder consumer to the Cabin Crew Ground activities, Details, and Schedules APIs, ensuring seamless data retrieval.

Tata Consultancy Services (TCS) | Remote *Network & DevOps Trainee (Apple Infrastructure Project)* | Oct 2025

- Troubleshot deep Linux OS and networking issues, analysing **system calls (strace), inodes**, and network traffic using tcpdump and Wireshark.
- Deployed and configured Nginx reverse proxies and load balancers, successfully resolving HTTP 403/404 errors and DNS routing failures.
- Hardened server security by implementing strict file permissions, writing custom Bash scripts for system automation, and configuring network firewalls (UFW, iptables, AWS Security Groups).

SYSTEMS ENGINEERING & AUTOMATION PROJECTS

Next-Generation Infrastructure Architecture (Post-M2 Decommissioning)

- Actively collaborating with cross-functional development and DevOps teams to architect and provision a completely new cloud infrastructure following the decommissioning of the legacy Mercury 2 (M2) environment.
- Participating in daily system design Knowledge Transfer (KT) sessions to define scalable networking, security, and deployment standards for the greenfield environment.

Cloud Cost Optimization & Observability Investigation

- Spearheaded an extensive AWS cost anomaly investigation, utilizing CloudTrail and Datadog to identify operational inefficiencies across Lambda, SNS, and SQS services.
- Analyzed system behaviors and resource allocation to provide actionable insights for reducing cloud spend and optimizing infrastructure performance.

Data Pipeline & Release Automation (GSE Events)

- Engineered a Python and Boto3-based automation solution to replace manual post-deployment checklists for the GSE Events Data Product, significantly accelerating release cycles and reducing human error.
- Automated data extraction workflows bridging legacy Oracle SQL databases with AWS S3 cloud storage, ensuring reliable data ingestion for downstream analytics.

Lightweight Python Web Server (OSI Layer 4/7):

- **Bypassed standard web frameworks to build a raw HTTP server from scratch using basic Python TCP Socket Programming, handling low-level GET/POST request parsing.**

Network Security & Packet Forensics:

- **Engineered a Python-based wireless penetration testing system to scan 802.11 traffic, capture handshakes, and test access point vulnerabilities.**

EDUCATION

Academy of Technology Engineering College | Hooghly, West Bengal *Bachelor of Technology in Mechanical Engineering* | Sep 2021 - 2025

- Applied mechanical systems-design principles and analytical problem-solving skills to complex cloud infrastructure and software architecture.

CERTIFICATIONS & QUALIFICATIONS

- AWS Certified Cloud Practitioner **(Perfect Score: 1000/1000)**
- Microsoft Certified: Azure Network Engineer Associate
- Google Cloud Certified Professional
- Oracle Cloud Infrastructure 2024 Certified Foundations Associate
- GitHub Certifications: Advanced Security, Actions, and Administration
- Postman Student Expert
- NASA Open Science 101 Qualification